

SELENIUM AUTOMATION ASSIGNMENT

Assignment 1 – The Multi-Locator Challenge

Execution Report • Selenium with Python • SauceDemo

Item	Details
Assignment	1 – The Multi-Locator Challenge
Application Under Test	SauceDemo Login
Automation Tool	Selenium WebDriver
Programming Language	Python
Browser	Google Chrome
Validation	URL contains /inventory.html
Execution Status	PASSED

This report documents the implementation and execution of the assignment, including the locator strategies used, validation logic, source code, and execution evidence.

1. Objective

The objective of this assignment is to automate the login process of a web application using Selenium WebDriver and demonstrate the use of multiple Selenium locator strategies within a single test scenario.

The test must locate the username field using By.ID, locate the password field using By.NAME, locate the login button using By.XPATH, perform the login, and verify that the resulting URL contains /inventory.html.

2. Assignment Requirements

No.	Requirement	Implementation
1	Open login page	Navigate to SauceDemo
2	Locate username	By.ID
3	Locate password	By.NAME
4	Locate login button	By.XPATH
5	Perform login	Enter valid credentials and click Login
6	Validate result	Assert /inventory.html is present in current URL

3. Test Environment

Component	Configuration
Operating System	Windows
IDE	Visual Studio Code
Language	Python 3.14 (64-bit)
Automation Framework	Selenium WebDriver
Browser	Google Chrome
Website	SauceDemo

4. Implementation Approach

Step 1: Launch Chrome using Selenium WebDriver.

Step 2: Navigate to the SauceDemo login page.

Step 3: Maximize the browser window.

Step 4: Find the username input using By.ID and enter standard_user.

Step 5: Find the password input using By.NAME and enter secret_sauce.

Step 6: Find the Login button using By.XPATH and click it.

Step 7: Check the current browser URL using an assertion.

Step 8: Print the execution result and current URL, then close the browser.

5. Locator Strategy Used

Web Element	Locator Strategy	Locator Used	Purpose
Username	By.ID	user-name	Enter username
Password	By.NAME	password	Enter password
Login Button	By.XPATH	//input[@id='login-button']	Submit login form

6. Test Script

```

from selenium import webdriver
from selenium.webdriver.common.by import By
import time

# Launch Chrome browser
driver = webdriver.Chrome()

# Open SauceDemo login page
driver.get("https://www.saucedemo.com/")

# Maximize browser
driver.maximize_window()

# Enter username using By.ID
username = driver.find_element(By.ID, "user-name")
username.send_keys("standard_user")

# Enter password using By.NAME
password = driver.find_element(By.NAME, "password")
password.send_keys("secret_sauce")

# Click Login button using By.XPATH
login_button = driver.find_element(By.XPATH, "//input[@id='login-button']")
login_button.click()

# Validate that URL contains /inventory.html
assert "/inventory.html" in driver.current_url

print("Assignment 1 Passed!")
print("Current URL:", driver.current_url)

time.sleep(3)

# Close browser
driver.quit()

```

7. Execution Evidence

The following screenshots were captured during the completed execution and are included as practical proof of the test run.

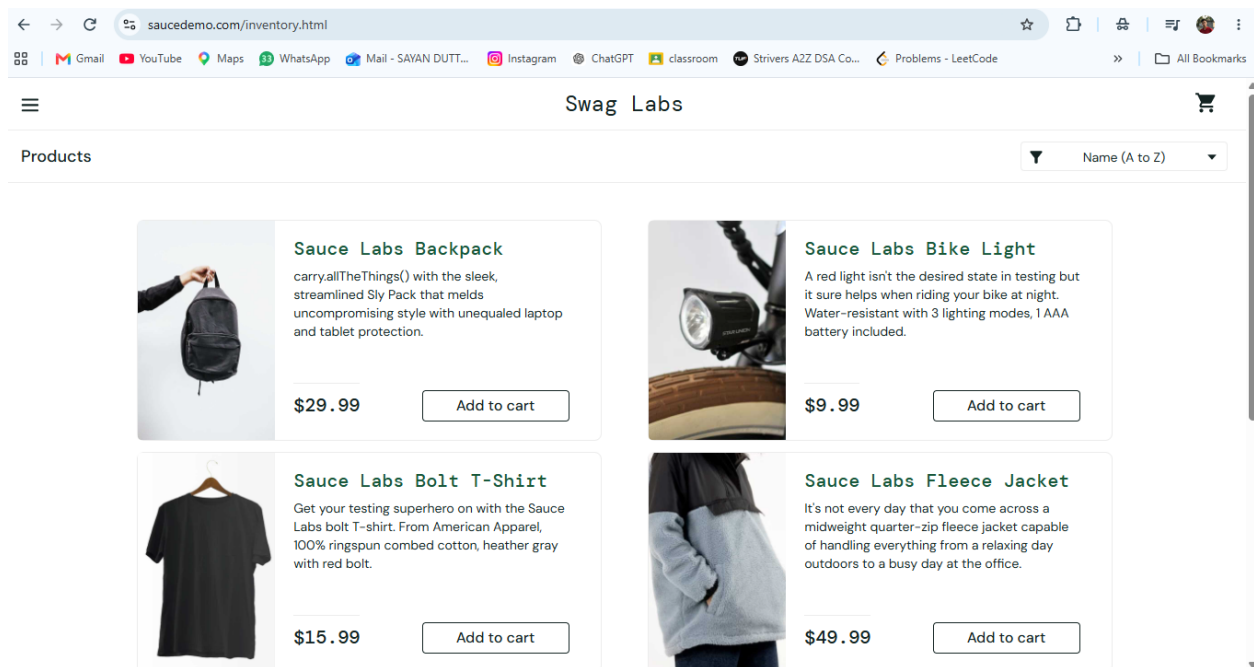


Figure 1 – SauceDemo login page before/at the start of the test.

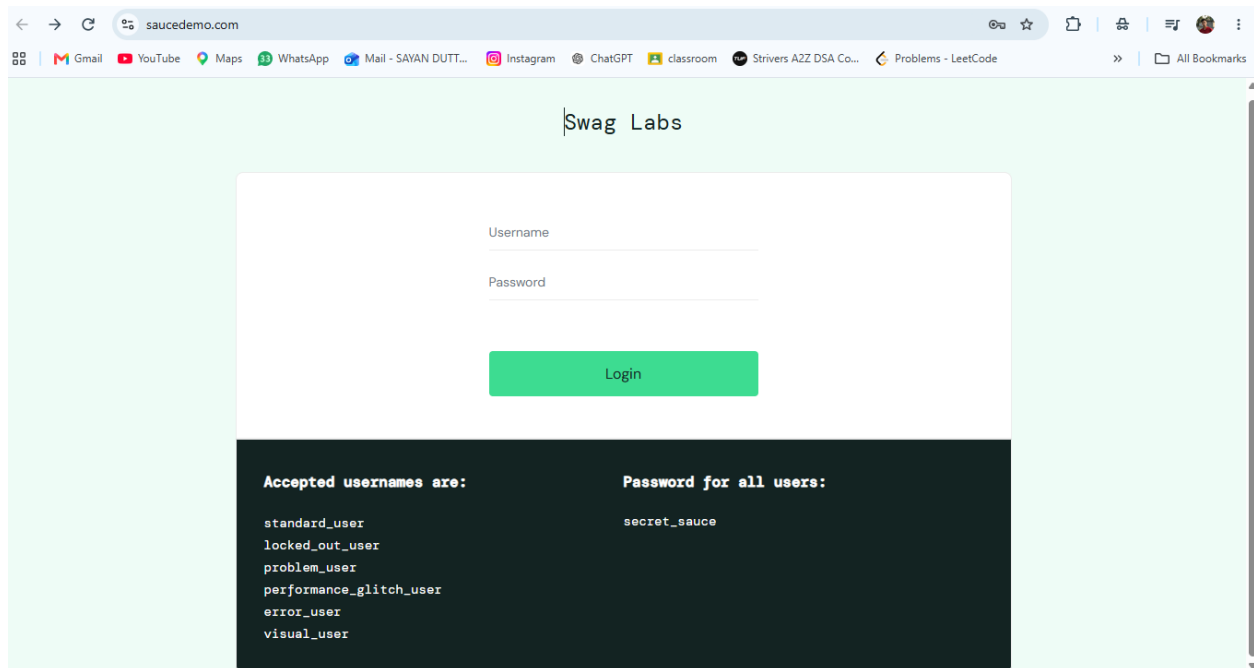


Figure 2 – SauceDemo inventory page reached after successful login.

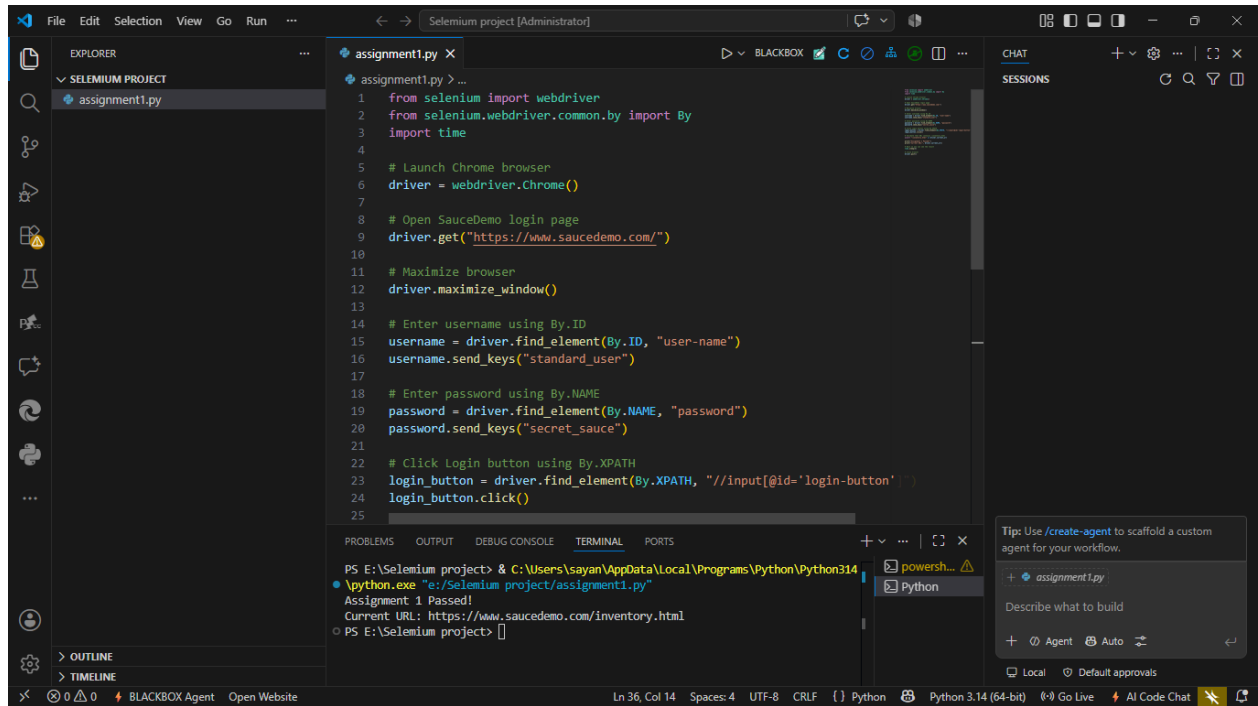


Figure 3 – VS Code terminal showing the passed assertion and resulting URL.

8. Test Result

Validation Point	Expected	Actual	Status
Login page opened	SauceDemo login page	SauceDemo login page	PASS
Username locator	By.ID	By.ID	PASS
Password locator	By.NAME	By.NAME	PASS
Login button locator	By.XPATH	By.XPATH	PASS
Post-login URL	Contains /inventory.html	https://www.saucedemo.com/inventory.html	PASS
Overall execution	All validations pass	All validations passed	PASS

9. Conclusion

The Multi-Locator Challenge was executed successfully using Selenium WebDriver with Python. The test demonstrated three different locator strategies: By.ID for the username field, By.NAME for the password field, and By.XPATH for the login button. After submitting the credentials, the automated test verified that the browser navigated to a URL containing /inventory.html. The assertion passed and the terminal displayed the successful execution result.

Final Status: PASSED

End of Report